SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I K. Mohammed Jasim *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	0	0

Total Marks 33 **/40**

Signature of the Faculty

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CO1 AT1

1. Given:-

- Original file size = 900MB
- compression = 40%.
- Encryption overhead = 5% of compressed size.

calculation of compressed file size:-

The file is compressed by 40%. meaning that 40% of the orginal file size is reduced.

Formula:

Compressed size = Orginal size x

$$= 900 \times \frac{60}{100}$$

Compressed size = 540 MB

calculation of encrypted file size.

$$\text{Overhead} = 540 \times \frac{5}{100}$$

$$= 27 MB$$

Encrypted file size:

Encrypted size = 567 MB

∴ The total encrypted file size is 567 MB

4. Calculation of total percentage reduction:-

Orginal file size = 900MB

Encrypted file size = 567MB

total reduction:

900 - 567 = 333 MB

percentage reduction:

$$\text{Reduction} = \frac{333}{900} \times 100$$

$$= 37\%$$

Hence, the final encrypted file is 37% smaller than the orginal file.

Role of the presentation layer:-

Compression: Reduces the amount of data transmitted over the network. It saved bandwidth, reduces transmision time, and improves network efficiency.

Encryption: coverts readable data into unreadable format to protect confidential information from unauthorized accen.

Data transmission: Converts data into a standard format that different computer system can understand.

Conclusion:-

The Presentation layer improves transmission efficiency through compression and provides security through encryption. In this example and 567 MB after encryption overhead.

7. FTP data Transfer and Application layer:-

Given:-

- Total data = 3GB
- Effective throughput = 120 Mbps.
- Size of each request = 3MB

For calculation, assume 1 GB = 1000 MB and ignore additional protocol overhead.

Calculate of total number of requests.

Number of requests = $\frac{\text{Total data}}{\text{request size}}$.

$= \frac{3 \times 1000}{3}$

$= \frac{3000}{3}$

Web services: HTTP and HTTPS allow user to access websites and web application.

User interaction: It provide an interface through which user access network service.

Conclusion;

The Application layer enables users to access network services and transfer file efficiently. In this Problem. transfer 3GB of data. requires 1000 requests with an estimate transmission time 200 seconds at 120 Mbps.

8. Given:-

Parameter	delay.
Application layer	4ms.
Transport layer.	8ms.
Network layer	5ms.
Data link layer	2ms.
Physical layer	1ms.
Transmission delay	12ms.

Calculation of total delay:

Formula:

$D_p = \Sigma$ Processing delays.

$D_p = 4 + 3 + 5 + 2 + 1$

$D_p = 15ms.$

∴ The total processing delay is 15 millisecond.

ehd to end delay:-

$D_{total} = D_p + D_t + D_{prop}$

D_p = Total processing delay.

D_t = Transmision delay.

D_{prop} = propagation delay.

$D_{total} = 15 + 12 + 18$

$= 45ms.$

Contribution of each OSI layer:-

1. Application Layer - 4ms.

The Application Layer provides network services to user and application. Protocol such as HTTP, FTP, and SMTP operates at this layer. It prepares application for communication.

3. Network layer - 5ms.

The Network layer determines the routes taken by packet from the sources to the destination. It uses logical addressing, such as IP addresses and routing decisions to forward packet across network.

Conclusion:-

The OSI layer work together to process, route, and transmits packet across a network. The total processing delay is 45 ms, excluding queuing delay and any other unspecified delays.

4. Frames, Protocol overhead and Transmission Efficiency:-

Given:

- Useful data = 80MB
- Total frame size = 1600 bytes
- Protocol overhead per frame = 80 bytes.

Assume 1MB = 1,000,000 bytes and that 1600 bytes represent the total frame size, including overhead.

Calculation of useful data per frame.

Each frame contain 1600 bytes of which 80 bytes are used for Protocol overhead.

Formula:-

Payload per frame = frame size - bytes.

= 1600 - 80

= 1520 bytes.

Calculation of total number of frames:-

80MB = 80×10^6 bytes.

= 80,000,000 bytes.

Number of Frames = $\frac{\text{Total useful}}{\text{Payload Per data}}$

= $\frac{80,000,000}{1520}$

= 52631.58

Calculation of total protocol. overhead

Each frame contain 80 bytes of overhead.

$= 52,632 \times 80$

$= 4,210,560$ bytes

$= \frac{4210,560}{1,000,000}$

Total overhead = 4.21056 MB

Calculate of total transmitted data.

Total Transmitted Data

$= 80 + 421056$

Calculate transmission efficiency.

Efficiency = $\frac{\text{Useful data}}{\text{Total Transmitted Data}}$

$= \frac{80}{84.21056} \times 100$

Efficiency $\approx 95\%$.